

Docker_Task4:

- Define two services in the **docker-compose.yml** file: **`web`** and **`redis`**.
- The **`web`** service:
 - Uses the local context (builds from the current directory).
 - Maps port **8000** on the host to port **5000** on the container.
- The **`redis`** service:
 - Uses the official **redis:alpine** image.
- Use this **Dockerfile** to define the container for the **web** service during the build process:

```
FROM python:3.7-alpine
WORKDIR /code
ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
RUN apk add --no-cache gcc musl-dev linux-headers
COPY requirements.txt requirements.txt
RUN pip install -r requirements.txt
EXPOSE 5000
COPY . .
CMD ["flask", "run"]
```

- Use **`app.py`** file that represents your Flask application:

```
import time
import redis
from flask import Flask
app = Flask(__name__)
cache = redis.Redis(host='redis', port=6379)
def get_hit_count():
    retries = 5
    while True:
        try:
            return cache.incr('hits')
        except redis.exceptions.ConnectionError as exc:
            if retries == 0:
                raise exc
            retries -= 1
            time.sleep(0.5)
@app.route('/')
def hello():
    count = get_hit_count()
    return 'Hello World! I have been seen {} times.\n'.format(count)
```